exotic material. Material that has a *negative* average energy density, as measured by someone moving through it at nearly the speed of light.

field. Something that is distributed continuously and smoothly in space. Examples are the electric field, the magnetic field, the curvature of spacetime, and a gravitational wave.

fission, nuclear. The breakup of a large atomic nucleus to form several smaller ones. The fission of uranium or plutonium nuclei is the source of the energy that drives the explosion of an atomic bomb, and fission is the energy source in nuclear reactors.

freely falling object. An object on which no forces act except gravity.

free particle. A particle on which no forces act; that is, a particle that moves solely under the influence of its own inertia. When gravity is present: A particle on which no forces act *except* gravity.

frequency. The rate at which a wave oscillates; that is, its number of cycles of oscillation per second.

frozen star. The name used for a black hole in the U.S.S.R. during the 1960s.

function. A mathematical expression that tells how one quantity, for example, the circumference of a black hole's horizon, depends on some other quantity, for example, the black hole's mass; in this example, the function is $C = 4\pi GM/c^2$, where C is the circumference, M is the mass, G is Newton's gravitation constant, and c is the speed of light.

fusion, nuclear. The merger of two small atomic nuclei to form a larger one. The Sun is kept hot and hydrogen bombs are driven by the fusion of hydrogen, deuterium, and tritium nuclei to form helium nuclei.

galaxy. A collection of between 1 billion and 1 trillion stars that all orbit around a common center. Galaxies are typically about 100,000 light-years in diameter.

gamma rays. Electromagnetic waves with extremely short wavelengths; see Figure P.2 on page 25.

Geiger counter. A simple instrument for detecting X-rays; also called a "proportional counter."

general relativity. Einstein's laws of physics in which gravity is described by a curvature of spacetime